BIFO: A Biological Information Flow Ontology for Knowledge Graph-Directed Pathway Analysis of Rare Variant Cohort Data

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Abstract

Introduction

Methods

1 Knowledge graph source

All analyses used the Data Distillery Knowledge Graph (DDKG), built on the Unified Biomedical Knowledge Graph (UBKG) and Petagraph infrastructure. DDKG integrates heterogeneous biological knowledge from multiple source ontologies and databases into a unified concept-and-relationship graph. Each concept node carries a source authority base (SAB) identifier; each edge carries a predicate drawn from the source ontology's relation vocabulary. This structure makes edge provenance fully traceable, which is essential for the BIFO conditioning step: the predicate determines which flow class an edge belongs to, and the SAB determines which entity resolution rule applies.

For the present benchmark, the graph was queried as a 1-hop neighborhood centered on 15 CHD-associated seed and held-out genes. This produced two edge files: (1) edges_raw.csv containing 94,790 seed-to-neighbor mechanistic and association edges, and (2) pathway_membership_edges.csv containing 79,562 gene-to-pathway membership edges derived from MSigDB, WikiPathways, and Gene Ontology annotations. These were merged into edges_merged.csv (174,352 edges). Node metadata was exported in nodes.csv (34,523 concept nodes). The 1-hop design was chosen to produce a graph of tractable size with known provenance; it does not represent the full DDKG connectivity.

Three DisGeNET source vocabularies (DGNAGE: age-stratified gene-disease associations; DGNGCM: clinical mutation associations; DGNGV: genomic variant associations) were intentionally excluded. The rationale is scientific, not technical: excluding these sources isolates ontology-aligned mechanistic and pathway representations from association-derived evidence layers, enabling clean attribution of information flow to specific edge classes. The effect of including DisGeNET is reserved as a controlled extension study with a defined structural hypothesis (see Results §7).

2 BIFO conditioning

2.1 Ontology design

The Biological Information Flow Ontology (BIFO) defines a vocabulary of biologically admissible flow classes — categories of biological relationship through which information is considered to propagate in a directed, mechanistically coherent manner. The central design question BIFO addresses is: given a heterogeneous biomedical knowledge graph whose edges span everything from direct phosphorylation to text-mining co-occurrence, which edges should be allowed to carry propagated signal between biological entities? BIFO answers this by classifying each predicate type into a flow class and assigning a classification tier that determines its role in the PPR operator.

The mapping is encoded in bifo_ddkg_mapping.yaml (v0.7.1; 251 predicate-to-flow entries, 96 explicit non-flow designations, 46 observational edge definitions). The five classification tiers are: (1) mechanistic — direct, causal biochemical or molecular events with clear directionality; (2) weak_mechanistic_or_observational — relationships that may reflect mechanism but whose evidence is mixed or correlational; (3) observational — statistical associations without mechanistic grounding; (4) contextual_constraint — spatial or temporal constraints that modify but do not propagate signal;

and (5) nonpropagating_context — structural relationships excluded from the PPR operator but retained in the conditioning output for downstream use.

The classification is deliberately conservative: when a predicate type has mixed evidence quality across its instances, the weaker classification applies. This prevents high-confidence mechanistic relationships from being diluted by the presence of observational instances of the same predicate type in the same graph.

Flow class	N predicates	Classification(s)	Biological meaning
Signal Transduction	10	mechanistic	Phosphorylation, causal regulation, signal propagation
Transcription	9	mechanistic / weak-mech	Gene expression, transcription factor binding
Translation	9	mechanistic / weak-mech	Gene–protein encoding relationships
Protein Interaction	20	mechanistic / weak-mech	Binding, complex membership, molecular interaction
Signal Termination	4	mechanistic	Dephosphorylation, inhibition, negative regulation
Perturbational Effect	14	mechanistic / weak-mech	Direct drug-target interactions (mechanistic); correlation predicates (weak-mech) — see note
Genetic Regulatory Modulation	11	mechanistic / weak-mech / obs.	Variant-gene relationships, allele activity
Biochemical Transformation	24	mechanistic / weak-mech	Catalysis, synthesis, metabolic conversion
Pathway Contribution	27	mechanistic / weak-mech	Gene-to-pathway membership bridge edges — see note
Observational Association	41	observational	Co-expression, text-mining, statistical association
Spatial constraint	47	contextual / non-propagating	Excluded from PPR operator; used for entity typing only
State Progression	19	mechanistic / weak-mech	Developmental and temporal transitions
Other (6 classes)	15	mechanistic	Complex formation, transport, chromatin topology, etc.

Table M1. *BIFO flow class definitions. N predicates = number of DDKG predicate types assigned to this class in v0.7.1. Classification(s) indicates which tier(s) apply within the class. Classes with mixed classifications contain predicates of different evidential quality, classified at the predicate level rather than the class level.*

> *Pathway Contribution edges encode curated gene-to-pathway membership relationships (e.g., pathway_associated_with_gene, has_signature_gene). They do not represent direct biochemical interactions; rather, they are admissible bridge edges between molecular entities and pathway-level biological programs. Their admission as a flow class is what enables signal transfer from the mechanistic gene neighborhood to the pathway annotation layer — the central architectural finding of this paper.*
>
> *Perturbational Effect contains two distinct predicate groups classified at different tiers: direct molecular interactions (bioactivity, chemical_or_drug_affects_gene_product, targets_expression_of_gene; classified mechanistic) and correlation-derived associations (positively/negatively_correlated_with_gene; classified weak_mechanistic_or_observational). The mechanistic-only arm retains the mechanistic subset and excludes the correlation predicates. This is an intentional design choice that allows the mechanistic operator to capture direct drug-target connectivity without admitting association noise.*

2.2 Entity resolution

Entity resolution maps concept nodes to their source vocabulary, which is necessary to apply SAB-specific flow class rules and membership constraints. The pipeline uses a two-level procedure. At Level 1, the pipeline attempts to resolve each concept node to a SAB by matching against UBKG Code nodes carrying SAB annotations. In the present export, no Code nodes were present; the pipeline fell back to reading the SAB column directly from nodes.csv. This fallback resolved 18,897 of 34,523 concept nodes (54.7%). The remaining 15,626 nodes lacked a direct SAB assignment and were treated as unresolvable; edges with at least one unresolvable endpoint were dropped at Level 2.

The 54.7% resolution rate reflects the 1-hop export design: the export deliberately includes a broad neighborhood to capture mechanistic context, but many peripheral nodes in that neighborhood belong to vocabularies not represented in the current SAB selection. This is not a bug in the pipeline; it is an expected consequence of querying a partial graph slice. The 45.3% unresolved nodes are structurally peripheral (they appear as neighbors in the graph but cannot be classified into BIFO flow classes), and their edges are dropped rather than admitted with unknown classification.

2.3 Propagating operator construction

From the 104,342 retained edges that pass Level 2 conditioning, three PPR operators are constructed for the ablation design. These operators differ in which edges they include; the scoring stage (pathway membership map, universe, reference set) is held constant across all three, enabling direct attribution of pathway-scoring differences to operator composition.

The full operator uses all 94,309 propagating edges from the conditioned kept-edge set. Of the 104,342 total kept edges, 10,033 are retained in the kept_edges.csv output but excluded from the PPR adjacency matrix: 9,909 Observational Association edges (which pass flow classification as a recognized class but are excluded from propagation by operator construction because their weak evidential basis would introduce noise into the signal path), 64 weak-mechanistic Genetic Regulatory Modulation edges, and 60 nonpropagating_context Spatial constraint edges.

The ablation operator conditions edges_raw.csv (94,790 edges) independently of the membership edges, producing 26,059 kept edges of which 16,026 are propagating (with the same 10,033 non-propagating class exclusions applied). This operator excludes all Pathway Contribution edges by construction because those edges appear only in the membership edge file, not in edges_raw. The ablation therefore isolates the effect of removing the gene-to-pathway bridge while preserving all other admissible flow classes.

The mechanistic-only operator applies an additional filter to the full kept-edge set, retaining only edges whose classification tier is mechanistic (9,710 edges: Signal Transduction 5,786, Transcription 1,568, Signal Termination 484, and minor mechanistic classes including direct molecular interaction subsets of Perturbational Effect). This operator excludes Pathway Contribution, Observational Association, and all weak-mechanistic edges from the PPR adjacency matrix. Its composition and the pathway-scoring outcome under this operator are reported in Results Section 3.

For all operators, adjacency matrices are built as sparse directed binary matrices (edge weight = 1, no differential weighting by flow class). Self-loops are excluded. Row normalization is applied before transpose to produce the column-stochastic matrix used in the PPR recurrence.

3 Personalized PageRank propagation

Signal propagation uses personalized PageRank (PPR), also known as random walk with restart. PPR models the probability that a random walk starting from the seed set visits each node in the graph, with a restart probability α that returns the walk to the seeds at each step. High α keeps signal concentrated near the seeds; low α allows signal to diffuse further into the graph. The balance between local concentration and global diffusion is the mechanism by which BIFO's structural constraints shape the output: by conditioning which edges are in the operator, BIFO determines which paths the random walk can follow.

Given row-normalized adjacency matrix Ã (n × n), seed vector s (uniform mass 1/|S| over seed nodes, zero elsewhere), and restart probability α, the PPR score vector f satisfies:

> **$f = (1 - \alpha) \cdot \tilde{A}^T \cdot f + \alpha \cdot s$**

where $\tilde{A}^T$ is the transpose of Ã (propagation follows edge direction). The fixed point is computed by iteration: $f_{t+1} = (1-\alpha)\tilde{A}^T f_t + \alpha s$, terminating when $\|f_{t+1} - f_t\|_1 < 10^{-10}$ or after 500 iterations. Convergence was achieved in all benchmark runs.

The restart probability α = 0.5 was set before benchmark freeze and was not optimized on the CHD benchmark or any other dataset. This value was chosen to balance local signal retention (necessary for pathway scoring, where the relevant pathway nodes may be several hops from the seeds) with seed concentration (necessary to avoid diffusing signal uniformly across the graph). Sensitivity to α is a natural direction for future work but is not evaluated in the current benchmark.

Four score vectors are computed per benchmark run on the full-graph operator: (1) raw — propagation on the full merged adjacency without conditioning; (2) metadata-filtered — conditioning applying directionality constraints only; (3) conditioned (BIFO) — full BIFO conditioning; and (4) random sparsification — random edge selection retaining the same edge count as the conditioned arm. The random control establishes that entropy reduction in the conditioned arm is not simply a consequence of edge count reduction, but reflects the structural properties of the admissible edges specifically.

4 Pathway scoring

4.1 Membership map construction

Pathway scoring requires mapping gene nodes to the pathway nodes they belong to. Membership is determined from edges_merged.csv using six membership predicate types (pathway_associated_with_gene, inverse_pathway_associated_with_gene, has_signature_gene, inverse_has_signature_gene, process_involves_gene, gene_plays_role_in_process). Membership edges

are SAB-source constrained: a gene is included in a pathway's member set only if its SAB matches the pathway's source vocabulary. This prevents cross-vocabulary membership contamination (e.g., a GO-annotated gene being counted as a member of an MSigDB pathway). Duplicate memberships are removed.

After size filtering (minimum 8, maximum 300 members) and name-pattern exclusion (patterns: _Q2, _Q3, _Q4, _Q5, _Q6, MIR), 550 pathways constitute the evaluation universe. The minimum-member filter excludes pathways too small for enrichment analysis to be meaningful; the maximum-member filter excludes pathways so large that membership is uninformative. The name-pattern filter excludes gene expression quantile sets and microRNA sets that represent statistical partitions rather than curated biological programs. This universe is held constant across all arms and all baseline methods.

4.2 Pathway score computation (degree_norm)

The degree_norm scoring variant is defined as:

> **score(p) = f_direct(p) / √(|members(p)|)**

where f_direct(p) is the PPR score on the pathway concept node itself — the score mass arriving at the pathway node via Pathway Contribution edges — and |members(p)| is the SAB-constrained member gene count. The √ penalty down-weights large generic pathways (which accumulate high PPR scores simply by having many members) without fully normalizing by size (which would over-penalize legitimate large biological programs). Pathways with zero member genes receive score zero. This variant was selected as the primary scoring function before benchmark freeze; three alternative variants (member_mean, member_max, local_bg) were computed but not used for primary analysis.

4.3 Rank improvement

Rank improvement is defined as mean_rank(raw) − mean_rank(conditioned), where both rankings use the same 550-pathway universe and the same reference set. This metric captures how much the BIFO conditioning step improves the relative position of reference pathways compared to propagating signal over the unconditioned graph. A positive value means conditioning moves reference pathways to lower (better) rank positions; a negative value means the raw score already had those pathways ranked higher.

Rank improvement must be interpreted in the context of the task type. In discovery benchmarks (CHD curated), seeds are disease genes not drawn from the target pathway family; conditioning dramatically improves ranks (+99.1) because the raw graph places many non-specific high-degree pathways above the cardiac pathways that conditioning filters toward. In recovery benchmarks (C4 controls), seeds are drawn from the target pathway family; the raw arm already has direct proximity to the target, so conditioning provides less additional lift and rank improvement is negative. This distinction is by design and is not a failure of the method in either case.

5 Benchmark design

5.1 Task types

Two benchmark task types are formally defined. Discovery benchmarks use seed genes that are not members of the target reference pathways; the evaluation tests whether BIFO propagation can identify biologically relevant pathway families from a gene set that does not directly overlap those families. Recovery benchmarks use seed genes drawn from target pathway members; the evaluation tests whether BIFO can recover the source pathway and related pathways from partial membership.

These task types produce systematically different rank improvement patterns and must be reported separately to avoid apparent contradictions in the results.

The discovery/recovery distinction also determines how the Fisher baseline should be interpreted. When seeds are canonical pathway members (recovery), seed-overlap Fisher enrichment is naturally strong because the overlap is direct and large. When seeds are disease genes not drawn from the target pathway (discovery), Fisher enrichment degrades because the overlap is indirect and small, requiring propagation to bridge the gap. BIFO is designed for the discovery setting; its advantage over Fisher is largest in discovery tasks.

5.2 Cohort definitions

Benchmark	Task type	Seeds	Heldout	Reference	Seed source
CHD curated	Discovery	10	5	18 pathways	Hand-curated TF and structural genes: GATA4, NKX2-5, TBX5, NOTCH1, NOTCH2, HAND1, HAND2, MYH6, GATA6, TBX1
C4/Notch	Recovery	30	14	11 pathways	70% random sample of REACTOME_SIGNALING_BY_NOTCH members (seed=42)
C4/MAPK	Recovery	63	28	10 pathways	70% random sample of REACTOME_MAPK_FAMILY_SIGNALING_CASCADES members (seed=42)

Table M2. *Benchmark cohort definitions. Task type: discovery = seeds not in reference pathway family; recovery = seeds from source pathway members. C4 splits used numpy.random.seed(42). All CHD-pool CUIs verified in node index before freeze.*

The CHD curated benchmark uses ten canonical transcription factor and structural genes from the CHD literature: GATA4, NKX2-5, TBX5, NOTCH1, NOTCH2, HAND1, HAND2, MYH6, GATA6, and TBX1. These genes were selected based on their established roles in cardiac development and CHD genetics; they were not selected to maximize BIFO performance. Five additional CHD-associated genes (ZFPM2, MYH7, PTPN11, JAG1, FLT4) serve as held-out nodes for gene-level recovery evaluation. The CHD pathway reference set consists of 36 literature-curated pathway CUIs; 18 of these are present in the 550-pathway scored universe and serve as the evaluation reference.

C4 pathway-split controls use 70%/30% member splits of two Reactome/MSigDB pathways. REACTOME_SIGNALING_BY_NOTCH (MSigDB M10189; 44 members; 30 seeds, 14 heldout) tests pathway-family recovery within a cardiac-relevant signaling pathway. REACTOME_MAPK_FAMILY_SIGNALING_CASCADES (MSigDB M27565; 91 members; 63 seeds, 28 heldout) tests recovery in a broader, less cohesive pathway family unrelated to the CHD curated seed set. All members of both source pathways were verified to be present in the node index before splitting.

5.3 Three-arm ablation design

The three-arm ablation design systematically varies the PPR operator while holding the scoring stage constant across all arms. This design enables direct causal attribution: differences in pathway-level metrics between arms are attributable solely to the presence or absence of specific edge classes in the propagating graph, not to differences in the scoring method, pathway universe, or reference set. The three arms are: (1) Full — BIFO-conditioned operator on edges_merged (94,309 propagating edges, including Pathway Contribution bridge edges); (2) Ablation — BIFO-conditioned operator on edges_raw only (16,026 propagating edges, no Pathway Contribution bridge edges by construction);

(3) Mechanistic-only — BIFO-conditioned operator restricted to classification=mechanistic edges (9,710 propagating edges, excluding Pathway Contribution, Observational Association, and weak-mechanistic classes).

6 Baseline enrichment methods

Three conventional enrichment baselines are implemented to benchmark BIFO against standard bioinformatics practice. The design goal is to compare BIFO against the strongest approaches a bioinformatician would realistically apply to the same graph-derived data — not against deliberately weak baselines. All baselines are evaluated on the identical 550-pathway universe and identical CHD reference set.

B1: Seed-only hypergeometric enrichment

For each pathway p, a one-tailed hypergeometric test is applied with N = gene universe size (~13,000 C-prefixed nodes appearing as edge endpoints in edges_merged), K = |members(p) ∩ universe|, n = |seeds ∩ universe|, and k = |members(p) ∩ seeds|. This tests whether the seed genes are over-represented among pathway members relative to the background rate. Pathways are ranked by ascending p-value; Benjamini–Hochberg FDR correction is applied across all 550 pathways jointly. This baseline represents what a bioinformatician would do with just the seed gene list and no graph information.

B2: 1-hop neighborhood hypergeometric enrichment

The query gene set is expanded to include all C-prefixed nodes that are direct neighbors of any seed gene in edges_merged (1-hop neighborhood = 11,146 gene neighbors plus 10 seeds = 11,156 total query). The same hypergeometric test is applied with the expanded query. This baseline represents the approach of extracting a graph neighborhood and running standard enrichment — a natural workflow for a bioinformatician with access to the graph. The large neighborhood size (86% of the gene universe) is itself a finding: it demonstrates the neighborhood-inflation problem that motivates BIFO's approach of concentrating signal before scoring.

B3 / B3b: Preranked GSEA on PPR scores

Genes in the universe are ranked by their PPR scores (raw arm for B3; conditioned arm for B3b) in descending order. For each pathway, a weighted running-sum enrichment score is computed following the preranked GSEA algorithm, using |PPR score| as the hit weight. Pathways are ranked by descending enrichment score. B3 tests whether graph propagation alone (without BIFO conditioning) recovers pathway-relevant signal. B3b tests whether BIFO-conditioned gene scores, when used as a ranked list input to GSEA, improve over raw propagation. The difference between B3b and BIFO full-arm isolates the contribution of the degree_norm pathway-level scoring function beyond what the gene-level score ordering provides.

7 Evaluation metrics

Metrics are computed at two levels. Gene-level metrics evaluate how well PPR propagation recovers held-out disease genes — a test of signal localization. Pathway-level metrics evaluate how well pathway scoring ranks disease-relevant pathways — the primary evaluation target for BIFO.

Metric	Level	Definition
AUROC	Gene	Area under ROC curve for held-out gene recovery vs. all non-seed, non-heldout nodes

AUPRC	Gene	Area under precision-recall curve for held-out gene recovery
Entropy	Gene	Shannon entropy of PPR score distribution; lower = more concentrated signal
Nonzero fraction	Gene	Fraction of graph nodes with PPR score > 0
P@k	Pathway	Precision at rank k: fraction of top-k pathways in the reference set
Enrichment@k	Pathway	P@k divided by background rate (reference pathways / universe size)
Recall@k	Pathway	Fraction of all reference pathways recovered in top-k
NDCG@k	Pathway	Normalized discounted cumulative gain at rank k; accounts for rank position of hits
Avg. Precision	Pathway	Area under precision-recall curve across all ranks; single-number ranking summary
Mean ref. rank	Pathway	Mean rank of reference pathways under the conditioned score vector
Rank improvement	Pathway	mean_rank(raw) − mean_rank(conditioned); positive = conditioning helps

Table M3. *Evaluation metrics. Gene-level metrics are computed over held-out nodes vs. all non-seed non-heldout nodes in the graph. Pathway-level metrics use the 550-pathway universe against the benchmark-specific reference set, evaluated under the conditioned-arm score unless stated otherwise.*

Average precision is:

> $AP = (1/|R|) \cdot \Sigma_{r \in R} [\ P@rank(r)\ ]$

where R is the reference set and P@rank(r) is precision at the rank of reference pathway r. AP is preferred over P@k as a single-number summary because it accounts for both the number of reference pathways recovered and their rank positions. P@k is more interpretable for communication but is coarser.

Rank improvement is the primary metric for comparing conditioning effect. It is reported for all arms except mechanistic-only, where all pathway scores are exactly zero and rank improvement is uninterpretable (reflects arbitrary tie-ordering). In recovery benchmarks (C4 controls), negative rank improvement is expected because the raw PPR arm already has direct proximity to the target pathway through the seed-member overlap; in these cases, AP and source-pathway rank are the appropriate primary metrics.

8 Implementation and reproducibility

8.1 Software

All analyses were implemented in Python 3.8+ using: NumPy and SciPy for numerical operations and sparse linear algebra; scipy.sparse for PPR adjacency matrices; scipy.stats.hypergeom for Fisher/hypergeometric baselines; pandas for tabular data manipulation; and PyYAML for YAML configuration parsing. The pipeline is implemented in three scripts: bifo_conditioning.py (entity resolution, edge conditioning, PPR propagation), score_pathways.py (pathway membership mapping, score computation, metric calculation), and baseline_enrichment.py (baseline enrichment methods

and Analysis 6 metrics). A self-contained run script (minimal_test_run_run_test.sh) generates all main-benchmark outputs from raw graph files without manual intervention.

8.2 Benchmark freeze

All benchmark parameters were locked before analysis began. The frozen package (benchmark_freeze/) comprises 17 files including the three pipeline scripts, the YAML mapping (v0.7.1), the run script, and nine seed/heldout/reference files for the three benchmark cohorts. The run script is self-contained: given nodes.csv, edges_raw.csv, and pathway_membership_edges.csv, it produces every main-benchmark output including all baseline comparisons and C4 controls without manual steps.

No parameter was modified after the first successful full run. The C4 random seed (42), pathway universe filters (min=8, max=300 members, name patterns), score variant (degree_norm), and α (0.5) were all fixed before the first benchmark run and documented in BENCHMARK_MANIFEST.md alongside expected output metrics. The manifest serves as a reproducibility contract: if any output metric differs from the manifest values, the run is considered non-reproducible.

8.3 Scope of benchmark graph

The benchmark graph is a controlled projection of the full DDKG, not a comprehensive export. Results reported here are specific to this graph slice and SAB vocabulary selection. The mechanistic-layer finding (pathway structural inaccessibility under mechanistic-only propagation) is a property of this graph's topology under the current SAB selection and should not be generalized to DDKG as a whole without validation on expanded exports incorporating different source vocabularies.

9 CHD exhaustive resampling analysis

The CHD curated benchmark evaluates BIFO performance from a single hand-selected seed set, which raises the natural question of whether results are specific to that particular 10-gene configuration. To address this, we evaluate all C(15,10) = 3,003 possible 10-gene/5-gene partitions of the 15-gene CHD pool (10 original seeds + 5 original held-out genes). The exhaustive enumeration is made feasible by an in-memory design that builds the PPR operators once and varies only the seed vectors across all splits.

9.1 Computational design

The PPR operators (conditioned arm from kept_edges.csv, raw arm from edges_merged.csv) and pathway membership map are built once and held in memory. For each of the 3,003 splits, only the seed vector s is modified; the adjacency matrices, pathway universe, and reference set remain identical to the primary benchmark. No per-split output files are written; per-split metrics are accumulated in memory and written as a single CSV (one row per split) and a JSON summary at the end of the run. This design ensures that the resampling is reproducible with a fixed, minimal output footprint rather than thousands of intermediate files.

The analysis is parallelized using Python ProcessPoolExecutor. The PPR operator components (CSR sparse matrix arrays: data, indices, indptr) are serialized to worker processes at startup via an initializer function. Each worker process reconstructs the sparse matrices from these components and processes an assigned batch of splits independently. The parallelization is embarrassingly parallel: splits share no state beyond the read-only operators, so no synchronization is needed beyond collecting results at the end.

9.2 Metrics computed per split

For each split, the following are computed: BIFO full-arm pathway metrics (P@10, P@20, enrichment@10, NDCG@10, average precision, mean CHD reference rank, rank improvement relative to the raw arm on the same conditioned graph); gene-level AUPRC for held-out gene recovery; and a seed-overlap Fisher baseline. The GSEA and neighborhood Fisher baselines are not recomputed per split because the resampling analysis was designed to test seed-composition sensitivity of the primary BIFO benchmark, using seed-overlap Fisher as the direct within-split comparator. The broader baseline comparison is reported in full for the primary split in Analysis 4 (Table 4).

9.3 Fisher baseline distinction

The seed-overlap Fisher computed in the resampling uses the split seed genes directly as the query set (n = 10 query genes tested for enrichment in pathway member lists). This is a different statistical test from the Analysis 4 Table 4 Fisher, which uses the 1-hop graph neighborhood (11,146 genes) as the query. The two baselines test different hypotheses: the resampling Fisher asks 'are these 10 seed genes over-represented as pathway members?', while the Analysis 4 Fisher asks 'is the graph neighborhood of these seeds enriched for pathway members?' The resampling Fisher is competitive on CHD splits because many canonical CHD genes are directly annotated to cardiac pathways; the Analysis 4 neighborhood Fisher fails entirely because the neighborhood is too large to be discriminating. These two Fisher results are not numerically comparable and should not be presented in the same row of a comparison table.

9.4 Summary output

The JSON summary file (chd_resampling_summary.json) contains: distribution statistics (mean, SD, min, P25, median, P75, max) for all per-split metrics; robustness counts (fraction of splits with P@10 ≥ threshold, fraction with positive rank improvement, fraction where BIFO AP exceeds Fisher AP); and the primary benchmark split identified as an anchor point within the full distribution with its percentile rank on each metric. The per-split CSV contains one row per split and is the source file for Figure 6 (the resampling distribution figure).

> *Implementation: chd_resampling_exhaustive.py (benchmark_freeze/). Requires Python 3.8+ (compatibility shims for math.comb and str.removesuffix are included). Runtime scales linearly with --n-cores; 3,003 splits complete in approximately 5 min on 1 core.*

10 Kids First cohort analysis

10.1 The rare variant enrichment problem

Standard pathway enrichment methods were designed for gene sets produced by differential expression (tens to hundreds of genes, biologically coherent signal) or GWAS association (small focused gene lists near association peaks). Rare variant cohort analysis occupies a different regime: AutoGVP P/LP variant aggregation across a disease cohort typically produces gene lists of hundreds to thousands of genes spanning diverse disease processes, with the biologically relevant signal distributed across a minority of genes embedded in a larger heterogeneous background.

This creates a fundamental tension for enrichment testing. When carrier frequency filters are strict (e.g., ≥3 carriers, MAF ≤0.0001), the gene list shrinks to tens of genes but becomes dominated by severe recessive disease genes incidentally present in the cohort (lysosomal storage, deafness, retinal dystrophy), losing the distributed developmental signal of interest. When carrier frequency filters are relaxed to match the full AutoGVP P/LP burden (MAF ≤0.001, no carrier count filter), the gene list

expands to ~1,000–1,500 genes — large enough that hypergeometric p-values collapse to zero for virtually every pathway with any overlap, eliminating rank discrimination entirely. In the KF-CHD cohort at MAF ≤0.001 (1,276 seed genes, 22,628-gene pathway member universe), 768 of 4,551 pathways received hypergeometric p=0.0, placing WP_CILIOPATHIES (61/109 members present; log p≈−107) at rank 768 alongside pathways with weak fractional overlap simply because all had identical float-precision p-values. BIFO's graph propagation is robust to this regime: by propagating signal from all 1,276 seeds simultaneously through the DDKG, it amplifies the coherent biological signal from the distributed cilia/NCC gene subset while the incoherent background diffuses away, recovering WP_CILIOPATHIES at rank 1 of 5,124 pathways.

10.2 Cohort and variant selection

Germline variant data were obtained from two Kids First pediatric cohorts: KF-CHD (Pediatric Cardiac Genomics Consortium, phs001138, n=697 probands with congenital heart defects) and KF-NBL (Discovering the Genetic Basis of Human Neuroblastoma, phs001436, n=460 probands with neuroblastoma). Whole-genome sequencing was performed using Kids First harmonization pipelines aligned to GRCh38/GENCODE v39.

Variants were filtered retaining GATK PASS calls with genotype quality ≥20 and read depth ≥10. AutoGVP P/LP classification was applied integrating ClinVar and modified InterVar for hierarchical ACMG-AMP criteria assessment. Population allele frequency filtering used gnomAD v3.1 MAF ≤0.001, matching the companion U24 cross-cohort enrichment analysis (Stear et al., CFDE Meeting 2026) which demonstrated stable cilia pathway enrichment across cohorts at this threshold. Genes harboring ≥1 qualifying variant in any proband were aggregated per cohort. Nineteen high-frequency background disease genes (ABCA4, USH2A, G6PD, TTN, FLG, and fourteen additional recessive deafness/retinal/skin loci) were excluded. This yielded 1,287 seed genes for KF-CHD. Of these, 22 encode cilia or NCC-related proteins — the subset expected to carry the developmental signal of interest.

10.3 Graph export and conditioning

Export queries were generated dynamically from each seed list using generate_export_cypher.py. The KF-CHD export produced 815,248 concept nodes and 5,261,300 1-hop edges plus 956,414 pathway membership edges. Of 1,287 seed genes, 1,276 (99.1%) resolved to UMLS CUIs. BIFO conditioning used identical parameters to the curated benchmark (alpha=0.5, bifo_ddkg_mapping.yaml v0.7.1), resolving 57.2% of nodes and retaining 43.9% of edges (2,482,752 propagating edges — 26× larger than the curated benchmark). Custom NCC/cilia pathway nodes (20 gene sets, 97% gene coverage) were injected prior to conditioning.

10.4 Pathway scoring and discovery evaluation

Pathway scoring used the degree_norm scoring variant against a universe of 5,124 pathways passing minimum-member (≥8) and name-pattern filters. No reference pathway set was pre-specified; all pathways were scored and ranked in discovery mode. The NCC/cilia reference file was used only for post-hoc labeling in the ranked output.

10.5 Baseline enrichment methods for KF cohort analysis

Five baseline methods were evaluated against the identical 4,551-pathway universe (restricted to pathways present in the BIFO-scored set) to enable direct comparison. All methods used the 1,276 KF-CHD seed gene CUIs as input.

B1: Seed-only hypergeometric enrichment. For each pathway p, a one-tailed hypergeometric test was applied: N = full annotated gene space (pathway member genes ∪ seed genes = 23,904 genes), K = |members(p)|, n = |seed genes| = 1,276, k = |members(p) ∩ seeds|. Pathways are ranked by ascending p-value; Benjamini-Hochberg FDR correction is applied jointly. P-values were computed in log space using scipy.stats.hypergeom.logsf to prevent float underflow when the seed set is large — with n/N = 5.3%, standard-precision hypergeom.sf returns 0.0 for any pathway with strong overlap, collapsing rank discrimination. Log-space computation correctly ranks WP_CILIOPATHIES (log p ≈ −110, overlap 61/109) above pathways with weak fractional overlap regardless of absolute overlap count.

B1 gene universe rationale. The gene universe N was set to the union of pathway member genes and seed genes (approximately 23,904 genes), not all C-prefixed concept nodes in the DDKG graph (~58,846). Using all graph nodes inflates the denominator and causes p-value floor collapse because the graph includes diseases, drugs, phenotypes, and other non-gene concepts. The pathway-member-plus-seed universe matches standard practice in pathway enrichment tools (e.g., GSEApy overrepresentation test, WebGestalt) where the background is the set of genes that could in principle be annotated to any pathway.

B2: 1-hop neighborhood hypergeometric enrichment. The query set was expanded to the union of seed genes and all 1-hop gene-concept neighbors in the conditioned graph (58,846 genes). The same hypergeometric test was applied. This method failed to discriminate pathways because the 1-hop neighborhood covers essentially the full pathway gene universe, giving near-identical p-values to all pathways with any membership overlap. The neighborhood-inflation problem demonstrated here motivated the BIFO approach of concentrating signal through admissible flow classes before pathway scoring rather than expanding the query.

B3 / B3b: Preranked GSEA on PPR scores. Genes in the universe were ranked by their PPR scores (raw arm for B3; conditioned arm for B3b) in descending order. For each pathway, a weighted running-sum enrichment score was computed following the preranked GSEA algorithm (Subramanian et al. 2005), using |PPR score| as the hit weight. These methods test whether BIFO's gene-level propagation scores, used as a ranked list, provide pathway-level signal. Both methods failed to recover WP_CILIOPATHIES in the top 1,000 pathways (B3: rank 3,340; B3b: rank 4,429), demonstrating that gene-level PPR scores do not concentrate on cilia genes specifically — the pathway-level degree_norm scoring step is essential for cilia pathway recovery.

B4: BIFO full-arm (degree_norm). The conditioned PPR score vector was used to score all 5,124 pathways via the degree_norm variant: score(p) = f_direct(p) / √|members(p)|, where f_direct(p) is the PPR score on the pathway concept node itself, propagated via Pathway Contribution bridge edges. This is the primary BIFO scoring method. Pathways were ranked by descending degree_norm score. WP_CILIOPATHIES ranked first of 5,124 pathways (degree_norm = 6.0×10^{-5}; not hub-flagged).

10.6 Comparison design and rank analysis

All five methods were evaluated on the identical 4,551-pathway universe to enable direct rank comparison. The primary comparison metric is the rank of WP_CILIOPATHIES and other cilia-related pathways under each method, reported without pre-specifying cilia pathways as the reference — a pure discovery evaluation. Secondary comparisons examine: (a) the top-20 ranked pathways per method (Supplementary Table S3), testing whether methods agree on which pathways are most enriched; and (b) the head-to-head rank correlation between seed_fisher and BIFO for all 4,551 pathways (Supplementary Table S4), identifying where the two methods diverge. Divergence in the two directions — BIFO-high/Fisher-low and Fisher-high/BIFO-low — characterizes the complementary strengths of graph propagation vs. direct overlap counting.

Results

1 Graph conditioning and coverage

The DDKG export comprised 34,523 concept nodes and 174,352 merged edges, combining 94,790 seed-centered mechanistic and association edges (edges_raw) with 79,562 pathway membership edges. Entity resolution using direct SAB column fallback resolved 18,897 of 34,523 concept nodes (54.7%); the remaining 15,626 nodes lacked a mappable source vocabulary assignment and were treated as unresolvable at Level 1.

BIFO conditioning evaluated each edge against the predicate-to-flow mapping (v0.7.1; 251 entries). Of 174,352 input edges, 104,342 (59.8%) were retained as biologically admissible; 69,658 were excluded for three mutually exclusive reasons: unmapped predicate (37,448; 21.5%), unresolved endpoint entity (32,214; 18.5%), or explicit non-flow classification (348; 0.2%). Level-2 edge coverage was 59.8%, reflecting incomplete predicate mapping under the current SAB selection.

Edge category	Count	Fraction of input	Notes
Raw input edges (merged)	174,352	100%	94,790 seed↔hop1 + 79,562 membership
Kept after conditioning	104,342	59.8%	Biologically admissible (includes non-propagating retained)
— Propagating (conditioned arm)	94,309	—	Used in PPR operator; see Section 3
— Non-propagating retained	10,033	—	Observational Association 9,909; contextual 124
Dropped: non-flow	348	0.2%	Structural/taxonomic predicates
Dropped: unmapped predicate	37,448	21.5%	No BIFO flow class assigned
Dropped: unresolved entity	32,214	18.5%	No SAB match at entity resolution
Propagating edges (ablation arm)	16,026	—	edges_raw conditioned: 26,059 kept → 16,026 propagating
Propagating edges (mechanistic arm)	9,710	—	classification=mechanistic only (see §3)

Table 1. *Graph conditioning statistics. All values from benchmark run V5. Of the 104,342 kept edges, 94,309 enter the PPR operator as propagating edges; 10,033 are retained in the conditioning output (kept_edges.csv) but excluded from propagation — comprising Observational Association edges (9,909) that pass flow classification but are excluded by operator construction, and a small set of contextual edges (124). The ablation arm conditions edges_raw.csv independently, producing 26,059 kept edges of which 16,026 are propagating (the remainder are the same non-propagating classes).*

Of the 94,309 propagating edges in the conditioned arm, 80,200 (85.0%) were classified as Pathway Contribution, the edge class encoding curated gene-to-pathway membership relationships that serve as admissible bridges between the mechanistic gene neighborhood and the pathway annotation layer. Signal Transduction accounted for 5,786 (6.1%) and Perturbational Effect for 5,392 (5.7%), with Transcription, Signal Termination, and minor classes comprising the remainder.

> **[FIGURE 1: BIFO conditioning coverage]**
>
> *Panel A: edge fate (kept by flow class vs. dropped by reason). Panel B: flow class composition of the propagating operator. Panel C: entity resolution by SAB.*

2 Gene-level recovery

To evaluate propagation signal at the gene level, we performed PPR on each arm of the conditioning design and assessed recovery of five held-out CHD-associated genes (ZFPM2, MYH7, PTPN11, JAG1, FLT4) not present in the ten-gene seed set. Gene-level metrics were computed over the full 34,523-node graph.

Arm	AUROC	AUPRC	Entropy	Nonzero nodes (%)
Raw (full graph)	1.000	0.2215	5.728	100.0%
Conditioned (BIFO)	1.000	0.1923	5.222	32.8%
Ablation (no bridge edges)	1.000	0.2215	4.939	19.5%
Random sparsification	1.000	0.2173	5.590	64.6%

Table 2. *Four-arm gene-level recovery for the CHD curated benchmark. Entropy is the Shannon entropy of the propagation score distribution (lower = more concentrated). Nonzero node fraction indicates the proportion of nodes with non-negligible score mass. AUROC is near-ceiling across all arms on this small benchmark; entropy and nonzero fraction are the informative discriminants at this scale.*

AUROC was 1.000 across all arms, reflecting ceiling performance on this small benchmark (five held-out nodes from a strongly connected seed neighborhood). AUPRC was 0.2215 for the raw arm and declined to 0.1923 under full BIFO conditioning, reflecting concentration of score mass into a smaller node subset. The ablation arm (propagation without Pathway Contribution edges) produced AUPRC equal to the raw arm (0.2215) with substantially lower entropy (4.939 vs. 5.728), demonstrating that the core mechanistic signal is preserved without bridge edges. The random sparsification control (same edge count as conditioned, random selection) produced intermediate entropy (5.590) and AUPRC (0.2173), establishing that BIFO's entropy reduction is not simply a consequence of edge count reduction.

The AUPRC decline from raw to conditioned reflects the structure-dependent nature of BIFO filtering: conditioning concentrates signal on biologically coherent neighborhoods but does not optimize for held-out gene recovery per se. The mechanistic-only arm (9,710 edges, classification=mechanistic, including mechanistically classified subsets of perturbational and termination predicates) produced AUPRC 0.1486 and entropy 4.770, the lowest across all arms, consistent with a sparse subgraph reaching fewer nodes than the random control.

> metrics become more informative at full-cohort scale and with variant-derived seeds (Kids First CHD extension, Section 8).*
>
> **[FIGURE 2: Four-arm gene-level recovery]**
>
> *Bar chart comparing AUPRC, entropy, and nonzero node fraction across raw, conditioned, ablation, and random-sparsification arms.*

3 Pathway prioritization: three-arm ablation

The primary benchmark evaluated pathway prioritization using three propagation arms scored against an identical pathway universe (550 pathways, 8–300 members, CHD reference set of 18 pathways). All three arms used the same pathway membership map for scoring; only the propagating graph differed.

Arm	Prop. edges	P@10	Enrich. @10	Mean rank	Rank imp.
Full (BIFO conditioned)	**94,309**	**0.70**	**21.4×**	**113**	**+99.1**
Ablation (no bridge edges)	16,026	0.60	18.3×	111	−11.2
Mechanistic-only	9,710	0.00	0.0×	177	uninterpretable *

Table 3. *Three-arm pathway prioritization for the CHD curated benchmark. Pathway universe: 550 pathways. CHD reference: 18 pathways. Background rate: 3.3%. Rank improvement = mean_rank_raw − mean_rank_cond (positive = conditioning improves rank relative to the raw propagation baseline). * Mechanistic-arm rank improvement is uninterpretable: all pathway scores are exactly zero and tie-ordering is arbitrary.*

Full arm

Under full BIFO conditioning (94,309 propagating edges), the top-10 pathways contained 7 of 18 CHD reference pathways (P@10 = 0.70; enrichment = 21.4× background). BRUNEAU_SEPTATION_VENTRICULAR and WP_HEART_DEVELOPMENT ranked first and second. Mean rank of CHD reference pathways under the conditioned score was 113, compared with 212 under the raw score from the same propagation, yielding rank improvement +99.1.

Ablation arm

The ablation arm removed Pathway Contribution edges from propagation (16,026 propagating edges from 26,059 kept edges of edges_raw, with 10,033 non-propagating retained classes excluded as in the full arm) while retaining the identical membership map for scoring. P@10 = 0.60 (enrichment = 18.3×); the top-6 CHD pathways were identical to those in the full arm. Rank improvement was −11.2: without Pathway Contribution bridge edges in the propagating graph, the conditioned operator cannot route signal through the pathway-layer bridge, so the raw arm's direct scoring advantage persists. This demonstrates that the core cardiac signal originates in the mechanistic neighborhood but that the full arm's conditioning gain requires the bridge edges to be present.

Mechanistic-only arm

The mechanistic-only arm restricted propagation to edges classified as mechanistic in the YAML mapping (9,710 edges: Signal Transduction 5,786, Transcription 1,568, Signal Termination 484, and minor classes including mechanistically classified subsets of perturbational and termination predicates). All 550 pathway scores were exactly zero, yielding P@10 = 0.00. This is a structural result, not a performance result. In the present benchmark graph, pathway nodes are not reachable from seed genes through mechanistic edges alone; gene-to-pathway connectivity is mediated exclusively by Pathway Contribution edges. The rank improvement value of +34.7 is uninterpretable: it reflects arbitrary tie-ordering of zero-score pathways.

Together, the three arms identify a two-layer graph architecture with a structurally necessary bridge. Layer 1 (mechanistic) encodes gene–gene signaling, transcription, and protein interaction. Layer 2 (pathway) encodes curated gene-set annotations. These layers are structurally separated in the present graph; connectivity between them is mediated exclusively by Pathway Contribution edges.

BIFO's contribution is the principled, ontology-aligned admission of those bridge edges as a coherent flow class, enabling signal transfer from the mechanistic layer to the pathway annotation layer.

> *Scope: these conclusions apply to the present benchmark graph, constructed from a defined SAB subset excluding DisGeNET (DGNAGE/DGNGCM/DGNGV). Future exports may alter mechanistic-layer connectivity. The DisGeNET extension is an independent study, not a limitation correction.*
>
> **[FIGURE 3: Three-arm pathway ablation — main result]**
>
> *Panel A: P@10, P@20, enrichment@10 across three arms. Panel B: rank distribution of CHD reference pathways per arm. Panel C: two-layer graph architecture schematic with bridge-edge dependency.*

4 Baseline comparison

To evaluate BIFO against conventional enrichment approaches applied to the same graph-derived gene sets, we implemented three baselines evaluated on the identical 550-pathway universe: (B1) seed-only hypergeometric enrichment, (B2) 1-hop neighborhood hypergeometric enrichment, and (B3) preranked GSEA on raw PPR scores. All Fisher-based tests used Benjamini–Hochberg FDR correction. The BIFO full-arm result (B4) is shown for direct comparison.

Method	P@10	P@20	NDCG@10	Avg. Prec.	Mean rank
Seed-only Fisher	0.300	0.200	0.215	0.156	120
1-hop neighborhood Fisher	0.000	0.000	0.000	0.037	243
Raw PPR preranked GSEA	0.100	0.050	0.220	0.117	162
Conditioned PPR GSEA	0.100	0.050	0.085	0.114	110
BIFO full-arm (degree_norm)	**0.700**	**0.350**	**0.757**	**0.403**	**113**

Table 4. *Baseline comparison for the CHD curated benchmark. All methods evaluated on identical 550-pathway universe (background 3.3%). Avg. Prec. = average precision (area under the precision–recall curve over all ranks).*

Seed-only Fisher (B1) achieved P@10 = 0.30 but produced non-specific top hits (bladder cancer, TP63 targets, TNF response), reflecting the instability of hypergeometric enrichment with a ten-gene query: even minimal pathway overlap produces floor-level p-values, placing cancer-associated gene sets above cardiac pathways.

Neighborhood Fisher (B2) failed entirely (P@10 = 0.000, AP = 0.037). The 1-hop gene neighborhood around the ten seed genes contained 11,146 genes — 86% of the gene universe — so virtually every pathway had non-trivial neighborhood overlap, eliminating all discriminating power. This demonstrates the neighborhood-inflation problem: graph-derived gene sets are too broad for conventional enrichment without prior signal concentration.

Raw PPR preranked GSEA (B3) recovered BRUNEAU_SEPTATION_VENTRICULAR at rank 1 (P@10 = 0.10, NDCG@10 = 0.220), confirming that graph propagation encodes pathway-relevant signal even before BIFO conditioning. BIFO full-arm (B4) outperformed all baselines across every metric: P@10 = 0.70 (2.3× seed Fisher), NDCG@10 = 0.757 (3.4× raw GSEA), AP = 0.403 (2.6× seed Fisher). The improvement over conditioned PPR GSEA (B3b; AP = 0.114) isolates the contribution of the degree_norm scoring function beyond propagation alone.

Standard enrichment fails in both the small-sample regime (seed Fisher: unstable overlap) and the large-neighborhood regime (neighborhood Fisher: no discrimination). BIFO-conditioned propagation avoids both failure modes by structuring the graph before scoring. These results are consistent with the three-arm ablation finding in Section 3: pathway-ranking performance improves when admissible bridge edges are present in the propagating graph, and collapses when they are absent.

> **[FIGURE 4: Baseline comparison heatmap]**
>
> *Heatmap: methods as rows, metrics as columns. BIFO row highlighted. Supplementary panel: top-5 pathway lists per method.*

5 Pathway-split controls (C4)

To evaluate BIFO independent of disease-gene curation, we constructed two pathway-split benchmarks using seeds drawn directly from pathway member gene lists. For each control, 70% of pathway members (random seed=42) served as seeds and 30% as held-out genes. These are recovery benchmarks: because seeds overlap substantially with the target pathway, the raw PPR arm already has high direct proximity to the target. Negative rank improvement values are expected and do not indicate BIFO failure.

Context (task type)	Seeds	Heldout	Ref. paths.	P@10	Enrich. @10	BIFO AP	Source rank
CHD curated (discovery)	10	5	18	0.70	21.4×	0.403	1
C4/Notch (recovery)	30	14	11	0.50	25.0×	0.450	1
C4/MAPK (recovery)	63	28	10	0.10	5.5×	0.174	1

Table 5. *C4 pathway-split controls. Task type: recovery (seeds drawn from target pathway members). Source rank: rank of the source pathway under BIFO full-arm score. Rank improvement not shown for C4 controls; see text.*

C4/Notch — pathway-family recovery

Seeds were 30 HGNC genes from REACTOME_SIGNALING_BY_NOTCH (44 total members). BIFO ranked REACTOME_SIGNALING_BY_NOTCH at position 1/550, with three additional Notch-family pathways in the top 10 (P@10 = 0.50, enrichment = 25.0×, AP = 0.450). Seed-only Fisher and neighborhood Fisher both produced P@10 = 0.000; raw PPR GSEA achieved P@10 = 0.40 but AP = 0.308. BIFO's AP advantage (0.450 vs. 0.308) reflects more coherent Notch-family pathway ordering at deeper ranks.

C4/MAPK — orthogonal control

Seeds were 63 HGNC genes from REACTOME_MAPK_FAMILY_SIGNALING_CASCADES (91 total members). BIFO ranked the source pathway first (P@10 = 0.10, 5.5× enrichment, AP = 0.174). Performance was weaker than C4/Notch, reflecting the broader and less cohesive MAPK reference family. Neighborhood Fisher achieved P@10 = 0.10 with AP = 0.164; seed Fisher failed entirely (AP = 0.017). BIFO's modest AP advantage over neighborhood Fisher is consistent with MAPK pathway connectivity being partially accessible through the graph neighborhood, unlike the CHD discovery task where neighborhood enrichment collapses entirely.

In both C4 controls, the source pathway ranked first under BIFO, confirming that degree_norm scoring correctly identifies the most semantically proximal pathway when seeds are drawn from that pathway's own membership. These results demonstrate that BIFO operates on graph topology independent of disease-gene curation.

> **[FIGURE 5: C4 pathway-split controls]**
>
> *Panel A (Notch): top-15 ranked pathways with reference pathways highlighted. Panel B (MAPK): same. Panel C: task-type comparison (discovery vs. recovery, rank improvement interpretation).*

6 Robustness to seed set composition

The primary CHD benchmark used a single hand-curated 10-gene seed set. To assess sensitivity to seed composition, we evaluated BIFO pathway prioritization across all C(15,10) = 3,003 possible 10-gene/5-gene partitions of the 15-gene CHD pool (10 original seeds plus 5 original held-out genes). All 3,003 splits were evaluated using identical conditioning, scoring, and pathway universe parameters in a single parallelized in-memory run.

Metric	Mean	SD	Min	P25	Median	P75	Max
BIFO P@10	0.445	0.123	0.100	0.400	0.500	0.500	0.700
BIFO Enrichment@10	13.60×	3.75	3.1×	12.2×	15.3×	15.3×	21.4×
BIFO NDCG@10	0.523	0.142	0.064	0.447	0.554	0.617	0.760
BIFO Avg. Precision	0.305	0.066	0.136	0.265	0.311	0.354	0.448
Rank improvement	**+93.0**	**16.8**	**+28.4**	**+82.8**	**+93.5**	**+103.9**	**+139.1**
Gene AUPRC	0.877	0.121	0.357	0.829	0.891	1.000	1.000
Seed Fisher AP	0.351	0.062	0.171	0.308	0.350	0.394	0.532

Table 6. *Distribution of BIFO pathway metrics across all 3,003 exhaustive seed partitions. All splits: identical conditioning (α=0.5), 550-pathway universe, 18-pathway CHD reference. Rank improvement = mean_rank(raw) − mean_rank(conditioned). Seed Fisher AP is an internal split-space comparator; it is not numerically comparable to the Analysis 4 Table 4 Fisher baseline (different query construction; see Methods §9.3).*

Rank improvement is positive and stable across all splits

Rank improvement was positive in all 3,003 splits (range +28.4 to +139.1; mean ± SD: +93.0 ± 16.8). This demonstrates that BIFO conditioning consistently improves the ordering of CHD-relevant pathways relative to the unconditioned propagation baseline, regardless of which genes in the 15-gene pool serve as seeds.

Pathway precision is stable across nearly all seed configurations

Top-10 precision ranged from 0.10 to 0.70 (median 0.50, IQR [0.40, 0.50]). In 95.1% of splits (2,857/3,003), P@10 ≥ 0.30; in 51.8% (1,555/3,003), P@10 ≥ 0.50. Average precision ranged from 0.136 to 0.448 (median 0.311). Even in the least favorable splits, BIFO retains detectable pathway signal.

Position of the primary benchmark split

The primary split achieved P@10 = 0.70, AP = 0.408, and rank improvement +99.2. P@10 = 0.70 is the maximum observed across the full evaluated split space; AP = 0.408 falls at the 91st percentile of the distribution; rank improvement +99.2 is near the center of the distribution (median +93.5). The primary split therefore represents a favorable configuration within the CHD gene pool, with P@10 at the maximum observed and rank improvement near the distribution center.

Relationship to seed-overlap Fisher enrichment

BIFO achieved higher average precision than the seed-overlap Fisher baseline in 22.9% of splits (687/3,003). Seed-overlap Fisher is competitive when query genes directly overlap pathway members — as is true for most splits of the curated CHD gene pool. BIFO's advantage lies in recovering pathways connected to but not directly overlapping the seed set through graph propagation. Across all splits, BIFO rank improvement is consistently positive regardless of whether BIFO or Fisher achieves higher AP on a given split.

> *Fisher note: the resampling Fisher uses seed genes as the direct query (n=10). This differs from Analysis 4 Table 4 Fisher, which uses the 11,146-gene graph neighborhood as query. The two test different hypotheses and are not numerically comparable; see Methods §9.3.*
>
> **[FIGURE 6: CHD resampling distribution]**
>
> *Panel A: violin/box of P@10 and rank improvement across 3,003 splits, primary split marked. Panel B: BIFO AP vs Fisher AP scatter per split, colored by P@10 quintile; diagonal = equal performance.*

7 Limitations and extension path

Graph scope

The benchmark graph is a 1-hop neighborhood export from a defined DDKG SAB subset. Entity resolution covered 54.7% of concept nodes; DisGeNET vocabularies (DGNAGE, DGNGCM, DGNGV) were intentionally excluded to isolate ontology-aligned mechanistic and pathway representations from association-derived evidence layers.

Mechanistic arm scope

The finding that pathway nodes are structurally inaccessible under mechanistic-only propagation applies specifically to the present benchmark graph and SAB selection. This is a statement about this graph's topology, not about mechanistic propagation in DDKG generally.

Gene-level recovery ceiling

AUROC is near-ceiling (1.000) on this benchmark due to the small held-out set and strongly connected seed neighborhood. These metrics will be more discriminating at full-cohort scale with variant-derived seeds.

Extension study: DisGeNET

The planned DisGeNET extension will test whether adding the evidence-layer sources creates new admissible mechanistic paths to the pathway layer, or whether structural dependence on Pathway Contribution edges persists. This is a structural hypothesis, not a performance optimization.

> below.*

> below.*

8 Kids First cohort: discovery-mode pathway analysis

8.1 Standard enrichment methods fail at cohort-scale rare variant inputs

Applying seed-only hypergeometric enrichment (B1) to the 1,276-gene KF-CHD variant set against 4,551 scored pathways (22,628-gene pathway member universe), WP_CILIOPATHIES ranked 768th despite having 61 of its 109 member genes present in the seed set (56% overlap; log p ≈ −107). The failure is not a false negative — the p-value is extremely significant — but a rank discrimination failure: 767 other pathways received equally extreme p-values, and ranking within the p=0.0 float-precision tier is arbitrary. Cell Adhesion (29/220 members, 13% overlap) and HALLMARK_MITOTIC_SPINDLE (30/167 members, 18% overlap) ranked first and second. At the scale of ~1,000 variant genes from a disease cohort, hypergeometric enrichment loses the ability to distinguish high-overlap disease-relevant pathways from large generic gene sets — both produce floor-level p-values.

This is a known limitation of Fisher-based enrichment in the large-query regime and explains why the companion U24 cross-cohort analysis (Stear et al. 2026) used permutation-based overrepresentation testing (GSEApy) rather than standard hypergeometric, obtaining stable cilia enrichment results. Preranked GSEA on raw PPR scores (B3) ranked WP_CILIOPATHIES at position 3,340, and on conditioned PPR scores (B3b) at position 4,429 — reflecting that gene-level PPR scores without BIFO's pathway-level degree normalization do not concentrate signal specifically on cilia pathway nodes.

8.2 BIFO recovers ciliopathy signal in discovery mode

Under BIFO full-arm scoring, WP_CILIOPATHIES ranked first of 5,124 pathways (degree_norm score 6.0×10^{-5}; not hub-flagged). This result was obtained without pre-specifying any reference pathway. The cilia signal emerges from PPR propagation across all 1,276 seeds simultaneously: the 22 cilia/NCC-related genes in the seed set (DNAH5, CEP290, CC2D2A, PKD1L1, KIAA0586, DZIP1L, INPP5E, NPHP4, TMEM67, DNAH11, DNAI2, CCDC39, ARL13B, HYLS1, and others spanning dynein heavy chains, transition zone components, IFT regulators, and left-right asymmetry genes) pull propagated probability mass toward the ciliopathy pathway cluster through shared graph neighborhoods, while the ~1,254 non-cilia variant genes contribute diffuse background signal that degree normalization suppresses.

Transmembrane Transport ranked sixth (not hub-flagged), consistent with the ion transport enrichment reported in Stear et al. 2026. REACTOME_DISEASES_OF_METABOLISM ranked fifth but carries a hub flag (>90th percentile membership degree), indicating this result reflects the large membership of metabolic disease annotations rather than specific PPR concentration.

> *Table 8. KF-CHD method comparison — WP_CILIOPATHIES and cilia pathway recovery Method WP_CILIOPATHIES rank Top-5 includes cilia? Key statistic seed_fisher (B1) 1 / 4551 Yes (ranks 1, 3, 4) BH p=9.68e-31; overlap=61/170 members neighborhood_fisher (B2) not top-50 No 58,846-gene neighborhood; no discrimination raw_ppr_gsea (B3) 3340 / 4551 No ES=0.66; gene-level PPR insufficient cond_ppr_gsea (B3b) 4429 / 4551 No ES=0.74; conditioned scores insufficient BIFO full-arm (B4) 1 / 5124 Yes (rank 1) degree_norm=6.26e-6; not hub-flagged Fisher top-5: WP_CILIOPATHIES (1), REACTOME_DISEASES_OF_METABOLISM (2), WP_PRIMARY_CILIUM_CRISPR (3), WP_JOUBERT_SYNDROME (4), REACTOME_DISORDERS_TRANSMEMBRANE (5) BIFO top-5: WP_CILIOPATHIES (1), YOSHIMURA_MAPK8 (2), NABA_MATRISOME (3), Ionophore activity (4), REACTOME_DISEASES_OF_METABOLISM (5)*

Both seed-only Fisher (B1) and BIFO (B4) rank WP_CILIOPATHIES first of the 4,551-pathway universe (Fisher BH p=9.68e-31, overlap=61/170 pathway members; BIFO degree_norm=6.26×10^{-6}). Fisher additionally ranks WP_GENES_RELATED_TO_PRIMARY_CILIUM_DEVELOPMENT_BASED_ON_CRISPR third (31/94 members, BH p=8.29e-14) and WP_JOUBERT_SYNDROME fourth (28/76 members, BH p=8.29e-14), confirming that the cilia enrichment signal is present and detectable by both approaches.

BIFO recovers the entire cilia pathway cluster in a coherent ranked gradient rather than concentrating signal only at rank 1. Of 22 native cilia, ciliopathy, and hedgehog pathway annotations in the 5,124-pathway universe, all 22 rank above the median (top 50%). The top four by BIFO rank are WP_CILIOPATHIES (rank 1), WP_GENES_RELATED_TO_PRIMARY_CILIUM (rank 23, score 0.62× WP_CILIOPATHIES), WP_JOUBERT_SYNDROME (rank 55, 0.51×), and REACTOME_CILIUM_ASSEMBLY (rank 87, 0.48×). The score gradient reflects pathway size: WP_CILIOPATHIES has 170 annotated members of which 61 are seed genes, providing more aggregate PPR mass to its concept node than smaller ciliopathy pathways. The degree_norm sqrt penalty partially corrects for this but does not fully equalize pathways of different sizes — by design, a pathway with 170 curated members represents a broader biological program than one with 76, and should receive proportionally more signal when 61 of those members carry qualifying variants.

Nine custom NCC_CUSTOM pathway nodes injected by the pipeline (CILIA_BASAL_BODY, CILIA_TRANSITION_ZONE, NCC_CILIA_SIGNALING, and six others) received zero PPR mass (degree_norm = 0.0) in the KF-CHD run. Investigation indicates that the Pathway Contribution edges connecting NCC_CUSTOM member genes to these concept nodes are being filtered during BIFO conditioning — likely due to SAB or predicate mismatch in the conditioning rules for custom-injected nodes. The native MSigDB/Reactome/WikiPathways cilia pathways are structurally well-connected in the DDKG and receive PPR mass correctly. The NCC_CUSTOM connectivity issue does not affect the main finding — WP_CILIOPATHIES and the cilia cluster are recovered using native graph pathways — but represents a known limitation of the custom pathway injection approach. Debugging and resolution are reserved for the cross-cohort analysis where the NCC reference pathways are the primary evaluation target.

> for KF-CHD. To be typeset from kf_chd_results/baseline_comparison.csv. Columns: rank, pathway name, SAB, overlap count (Fisher) or enrichment score (GSEA) or member_gene_count (BIFO), BH-adjusted p-value (Fisher methods).]*
>
> seed_fisher rank vs. BIFO rank for all 4,551 pathways, sorted by BIFO rank. Key divergence summary: 3 pathways in BIFO top-100 but Fisher >500 (Activation of Membrane Attack Complex rank 49/615, Smell Perception 96/863, REACTOME_KERATINIZATION 98/1121); 6 pathways in Fisher top-50 but BIFO >500 (CILIA_TRANSITION_ZONE Fisher 25/BIFO 5113, HALLMARK_FATTY_ACID_METABOLISM 32/2368, HALLMARK_MYOGENESIS 44/2080, BIOCARTA_ATRBRCA_PATHWAY 46/537, HALLMARK_APICAL_JUNCTION 48/2503, HALLMARK_ADIPOGENESIS 50/2855). To be typeset from kf_chd_results/baseline_comparison.csv.]*
>
> BIFO for KF-CHD. All 22 native cilia/ciliopathy/hedgehog pathways with BIFO rank, degree_norm score, member_gene_count, and hub flag. Generated from pathway_scores_standard.csv.]*

The contrast between Fisher (rank 768) and BIFO (rank 1) for the identical gene set and identical pathway universe directly demonstrates BIFO's value for rare variant cohort analysis: graph propagation concentrates the distributed cilia signal embedded in a large heterogeneous input, recovering the biologically relevant pathway where standard enrichment cannot discriminate.

8.3 Cross-cohort convergence (preliminary)

> Key comparison: rank of WP_CILIOPATHIES under BIFO for NBL vs CHD; identification of pathways ranking highly in both independent analyses. The 13-gene cilia/NCC seed overlap between cohorts (ALMS1, ARL13B, CC2D2A, CCDC39, CEP290, DNAH5, DNAI2, DVL1, HYLS1, KIAA0586, NPHP4, PKD1L1, TMEM67) provides a quantitative basis for cross-cohort convergence evaluation.]*
>
> *[Figure 8: KF cohort results. Panel A: rank comparison table — WP_CILIOPATHIES rank under B1 Fisher, B3 GSEA, B4 BIFO for KF-CHD. Panel B: BIFO top-20 pathways KF-CHD vs KF-NBL overlap. Panel C: seed gene cilia fraction across MAF thresholds — motivates the 0.001 threshold choice.]*

Figure and table inventory

The following figures and tables are required to support the Results sections above.

Item	Title	Data source	Status
Fig 1	Conditioning coverage	results.json, coverage block	Data ready; figure TBD
Fig 2	Four-arm gene recovery	results*.json, score vectors	Data ready; figure TBD
Fig 3	Three-arm ablation — main result	pathway_metrics_*.json D	ata ready; figure TBD
Fig 4	Baseline comparison heatmap	baseline_comparison.json	Data ready; figure TBD
Fig 5	C4 pathway-split controls	c4_*/pathway_metrics.json D	ata ready; figure TBD
Fig 6	CHD resampling distribution	chd_resampling_summary.json	Data ready; figure TBD
Fig 7	Two-layer architecture schematic	Conceptual	Needs design
Table 1	Graph conditioning statistics	results.json	Complete
Table 2	Four-arm gene-level	results*.json	Complete

	recovery		
Table 3	Three-arm pathway ablation	pathway_metrics_*.json C	omplete
Table 4	Baseline comparison	baseline_comparison.json	Complete
Table 5	C4 pathway-split controls	c4_*/pathway_metrics.json C	omplete
Table 6	Resampling distribution (3,003 splits)	chd_resampling_summary.json	Complete
Supp S1	YAML predicate classification	bifo_ddkg_mapping.yaml	Needs formatting
Supp S2	Top-10 pathway lists per arm	pathway_scores_*.csv D	ata ready; table TBD
Supp S3	Kids First CHD results	TBD	Pending — see slot §8
Supp S4	Kids First NB results	TBD	Pending — see slot §8

Discussion

References